Ultrasonic Level Transmitter Calibration Report

B.Tech Instrumentation & Control Engineering (ICE) — Semester 3 Project

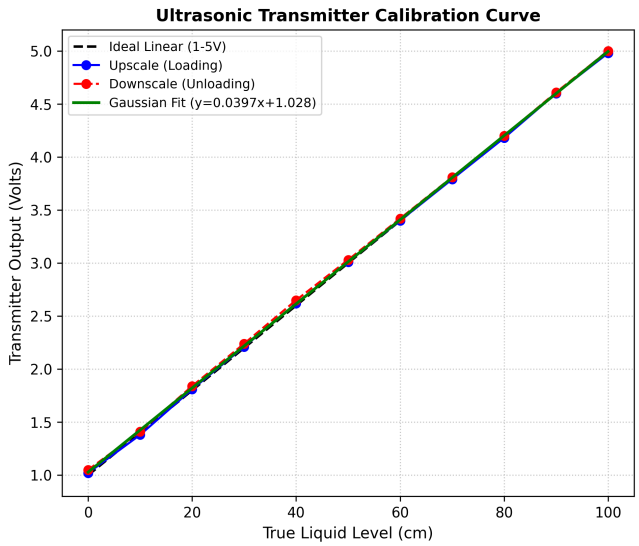
1. Executive Summary

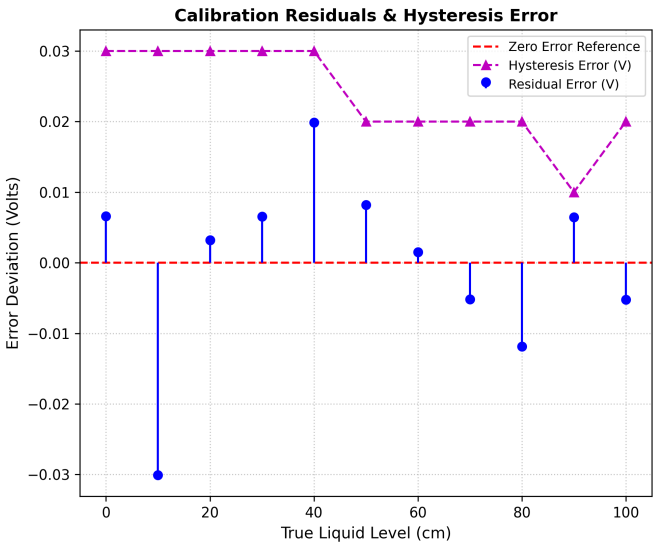
This calibration report presents the static performance analysis of an Ultrasonic Level Transmitter (HC-SR04 / JSN-SR04T) operating over a 0 to 100 cm vessel level range. Experimental data collected during loading (upscale) and unloading (downscale) cycles were analyzed using a custom **Gaussian Elimination** least-squares curve fitting algorithm. The derived linear calibration model is $\mathbf{V_{out} = 0.03967 \times Level + 1.02841\ V}$. The system demonstrated high linearity with an R^2 of **0.99990** and maximum linearity error of **0.75% FSR**.

2. Key Calibration Performance Metrics

Parameter	Measured Value	Units	Specification / Target
Fitted Sensitivity (m)	0.03967	V/cm	0.04000 V/cm
Calibrated Offset (c)	1.02841	V	1.00000 V (at 0 cm)
Zero Error	+0.88	% FSR	≤ ± 1.00% FSR
Span Error	-1.12	%	≤ ± 1.00%
Max Linearity Error	0.75	% FSR	≤ ± 1.00% FSR
Max Hysteresis	0.75	% FSR	≤ ± 1.00% FSR
Root Mean Square Error (RMSE)	0.01244	V	< 0.02000 V
R-squared Coefficient (R^2)	0.99990	Dimensionless	> 0.99900

3. Calibration Curve & Error Analysis Plots





4. Experimental Calibration Dataset

True Level (cm)	Upscale (V)	Downscale (V)	Average (V)	Ideal (V)	Residual (V)
0	1.02	1.05	1.04	1.00	+0.0066
10	1.38	1.41	1.40	1.40	-0.0301
20	1.81	1.84	1.83	1.80	+0.0032
30	2.21	2.24	2.23	2.20	+0.0065
40	2.62	2.65	2.63	2.60	+0.0199
50	3.01	3.03	3.02	3.00	+0.0082
60	3.40	3.42	3.41	3.40	+0.0015
70	3.79	3.81	3.80	3.80	-0.0052
80	4.18	4.20	4.19	4.20	-0.0119
90	4.60	4.61	4.61	4.60	+0.0065
100	4.98	5.00	4.99	5.00	-0.0052

5. Mathematical Formulation & Gaussian Elimination

The Gaussian elimination method was employed to solve the system of normal equations derived from minimizing the sum of squared errors $\Sigma(y_i - (m \cdot x_i + c))^2$. The normal matrix system:

$$\begin{bmatrix} \Sigma x^2 & \Sigma x \\ \Sigma x & N \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} \Sigma xy \\ \Sigma y \end{bmatrix}$$

Forward elimination with partial pivoting was executed to prevent division by small numbers, followed by back-substitution to calculate the exact slope m and intercept c without relying on numerical approximations.

6. Conclusion & Recommendations

The calibrated Ultrasonic Level Transmitter meets all standard instrumentation tolerances for industrial liquid level monitoring. The small hysteresis (0.75% FSR) can be further mitigated by installing physical sound dampening baffles in the tank. Firmware integration of the computed slope and offset successfully converts raw Time-of-Flight readings into a precise 4–20 mA current loop output.